

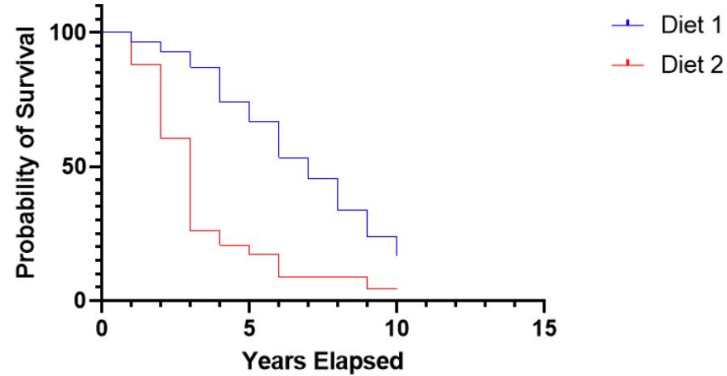
Survival Modeling

Instructor: Caryn Geady, PhD

Date: 2nd March 2026

What is Survival Analysis?

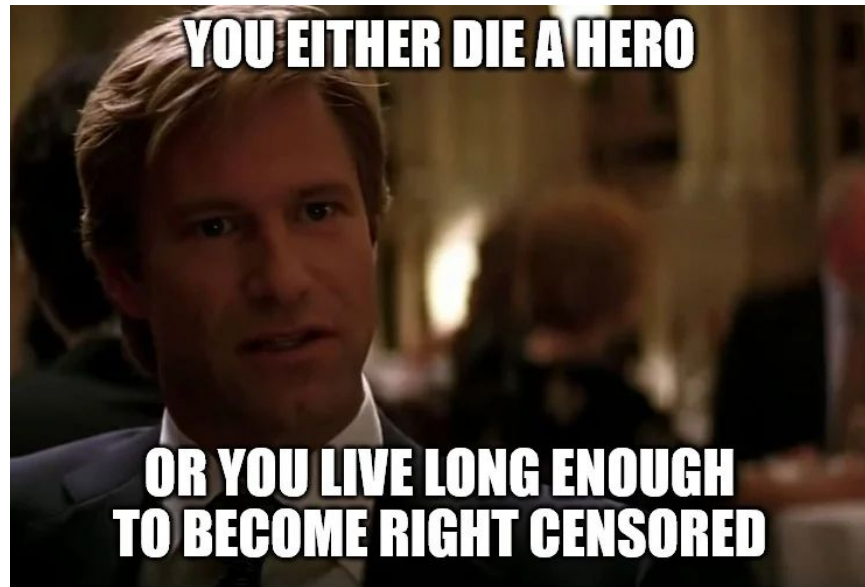
- A set of statistical approaches used to find out the **time** it takes for an **event of interest** to occur;



- **Units of time:** years, months, weeks, days, etc.
- **Event:** could be anything of interest (from dog diets to patient survival).

Today's Objectives

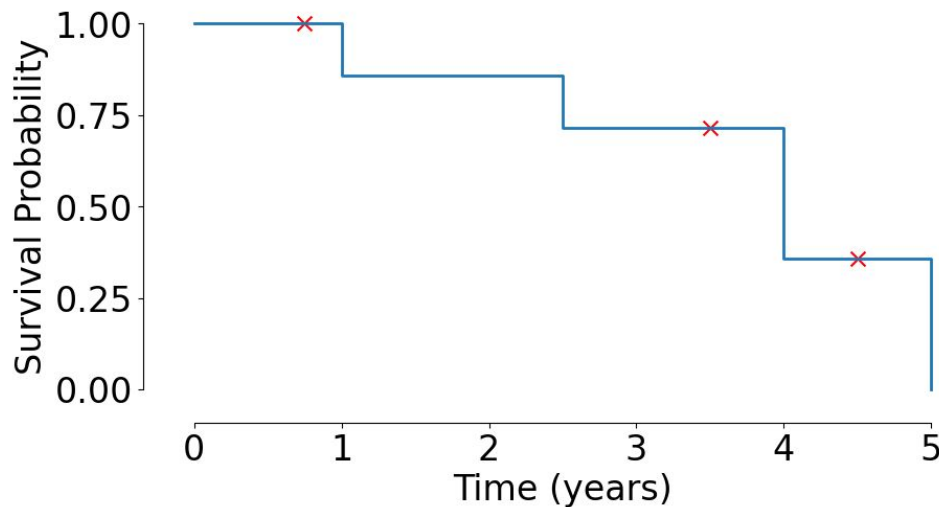
- Understand the concepts of **survival analysis** and **censoring**;
- **Key techniques:**
 - Kaplan-Meier estimator,
 - Cox Proportional Hazards model,
- **Beyond statistics:**
 - Regularized Cox models,
 - Random Survival Forests,
- Use Cases/Applications;
- **Metrics for evaluation.**



Key Techniques

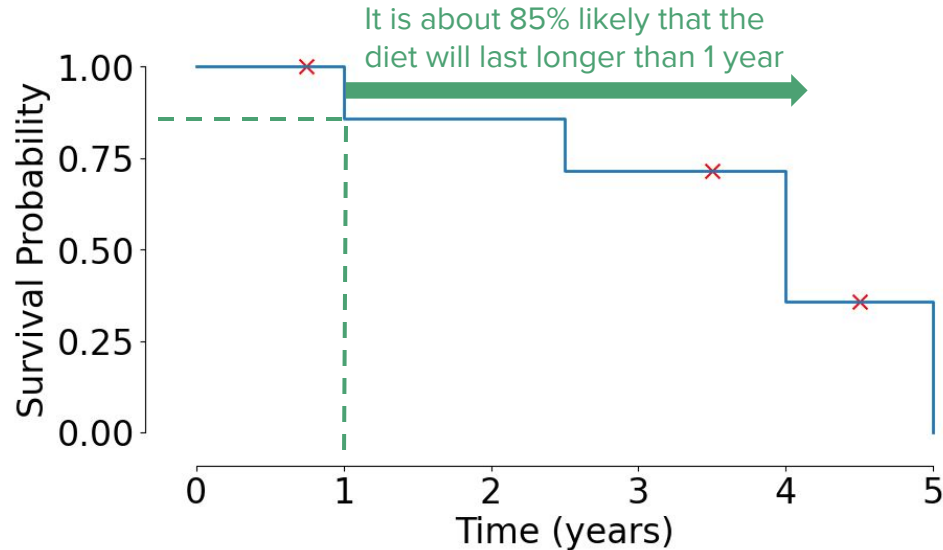
Kaplan-Meier Estimator: What It Is

- A non-parametric method used to estimate **survival probabilities**;
- Visualized via a **step function** survival curve with components:
 - Survival probability;



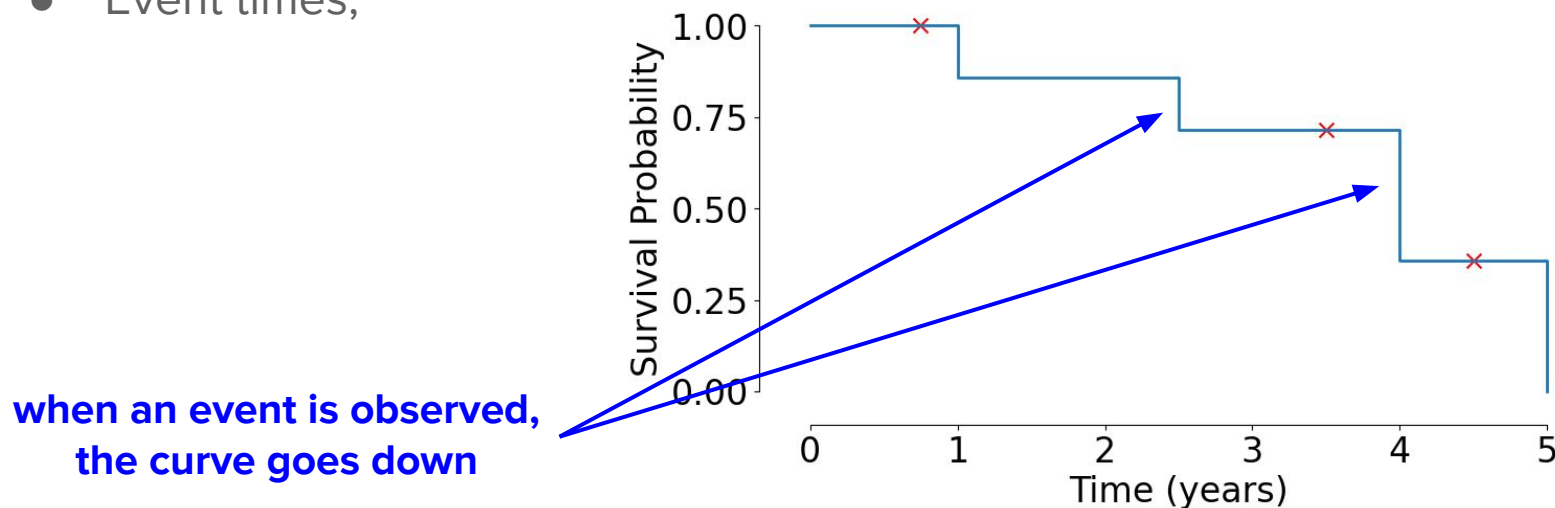
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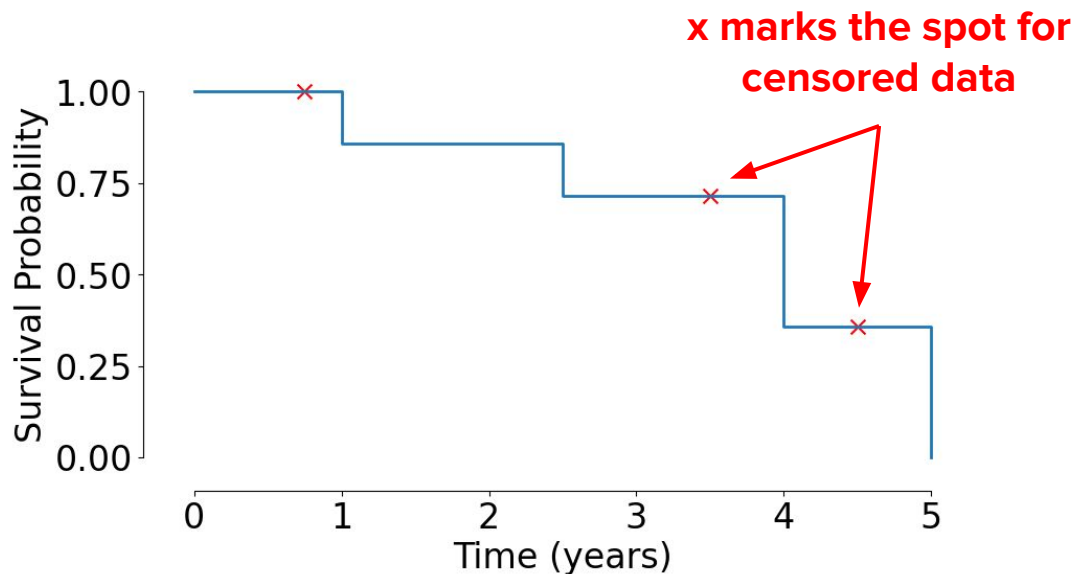
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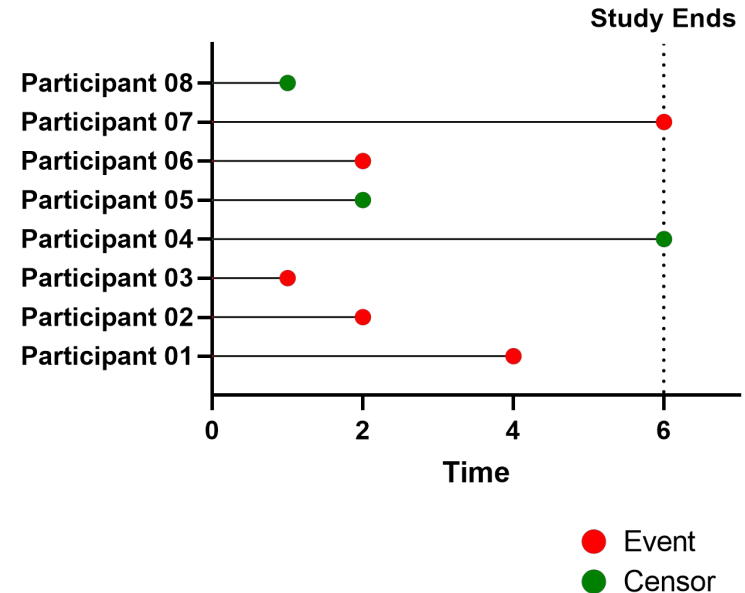
Kaplan-Meier Estimator: What It Is

- A non-parametric method used to estimate survival probabilities;
- Visualized via a **step function** survival curve with components:
 - Survival probability;
 - Event times;
 - Censored data.



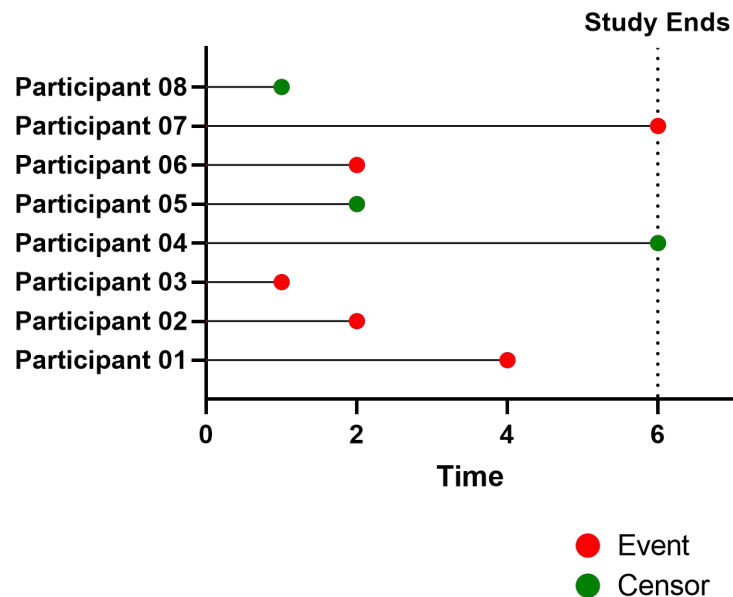
What is Censoring?

- Censoring occurs when we do **not observe the exact time** of an event.



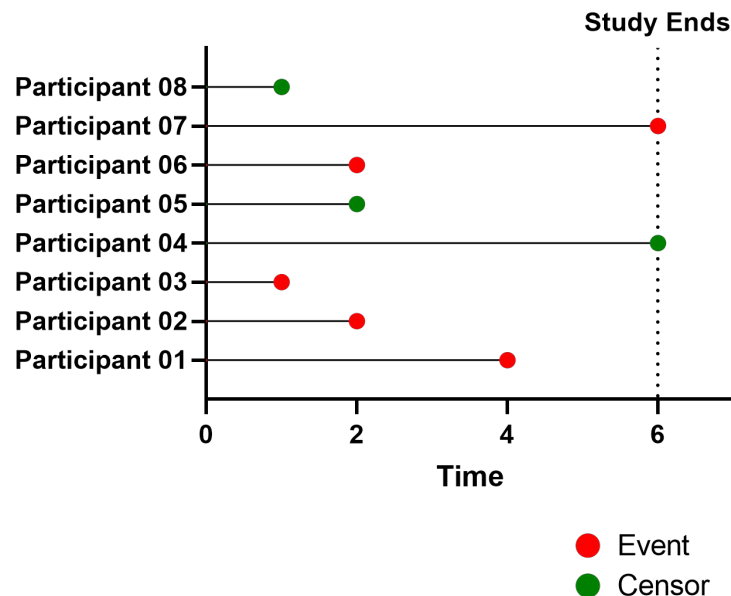
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 - **Right censoring:** The event hasn't occurred yet by the end of observation.



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- **Types of censoring:**
 - **Right censoring:** The event hasn't occurred yet by the end of observation.
 - **Left censoring:** The event occurred before observation began.
 - **Interval censoring:** The event occurred within a known time range.



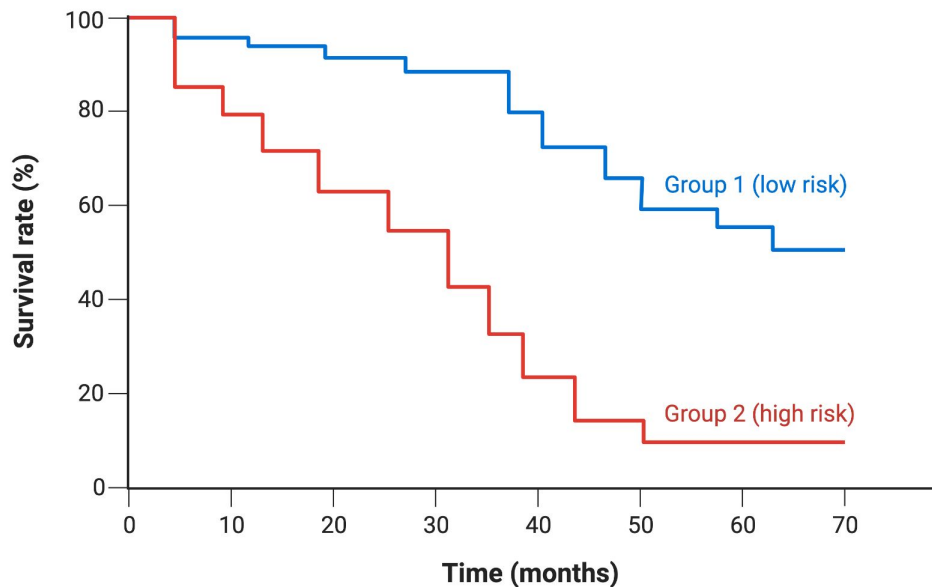
Interpreting Kaplan-Meier Curves

General

- Steeper slope → higher event rate;
- Flatter slope → lower event rate;

2+ Curves

- Parallel curves → similar survival experiences;
- Diverging curves → differences in survival between groups



Kaplan-Meier: The Good, the Bad and the Ugly

Advantages:

- Handles censored data effectively.
- Provides a clear visual of survival over time.
- Easy to compare multiple groups.

Limitations:

- Assumes censoring is unrelated to the outcome.
- Limited for small sample sizes or rare events.
- Does not model covariates directly (use Cox regression for this).

Cox Regression

- A statistical technique used to explore the relationship between the **survival time** of subjects and one or more **predictor variables**.

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- **Mathematical formulation:**

$$H(t) = H_0(t) \times \exp [b_1x_1 + b_2x_2 + \cdots ..b_kx_k]$$

Cox Regression

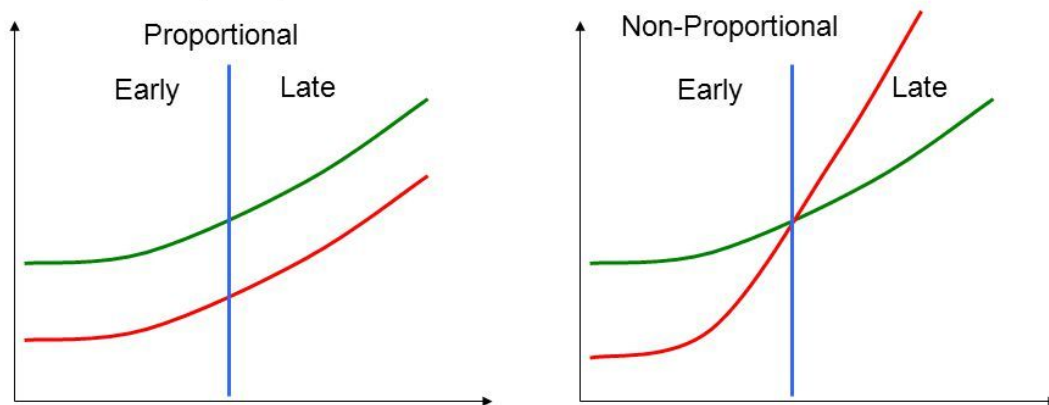
- A statistical technique used to explore the relationship between the **survival time** of subjects and one or more **predictor variables**.
- **Mathematical formulation:**

The diagram illustrates the mathematical formulation of the Cox Regression model. It features the equation $H(t) = H_0(t) \times \exp[b_1x_1 + b_2x_2 + \dots + b_kx_k]$. The components are visually linked as follows:

- Hazard Function:** A red box containing $H(t)$ is pointed to by a red arrow from the label "Hazard Function" below it.
- Baseline Hazard:** A green box containing $H_0(t)$ is pointed to by a green arrow from the label "Baseline Hazard" below it.
- Coefficients:** The term $\exp$ is followed by a sum of products $b_1x_1 + b_2x_2 + \dots + b_kx_k$. Each product is shown in a box split vertically, with the coefficient b_i in orange and the predictor variable x_i in blue. Three orange arrows point from the label "Coefficients" below to the orange b_i parts of the first three product boxes.
- Predictor Variables:** Three blue arrows point from a single point above to the blue x_i parts of the first three product boxes.

Cox Regression: Key Assumptions

- **Proportional Hazards:** The ratio of hazards between individuals is constant over time.
- **Linearity:** Log hazard is a linear function of the covariates.
- **Independence:** Survival times are independent given the covariates.



Cox Regression: Model Interpretation

- The exponentiated coefficient $\exp(\beta_i)$ represents the hazard ratio for a one-unit increase in X_i ;

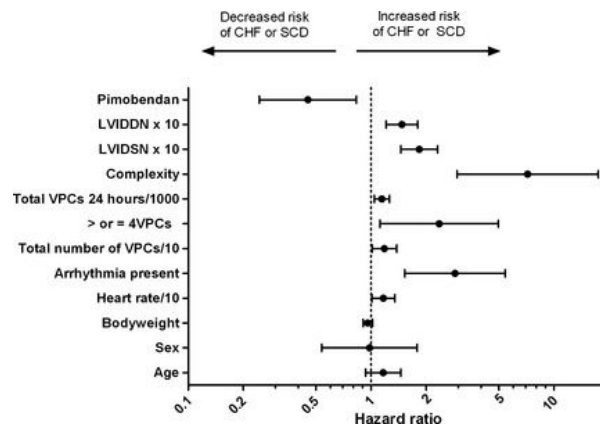
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$$H(t) = H_0(t) \times \exp[b_1x_1 + b_2x_2 + \dots + b_kx_k]$$

- A hazard ratio >1 indicates increased risk;
- A hazard ratio <1 indicates decreased risk.



Forest plot comparing hazard ratios for different predictor variables.

Beyond Statistics

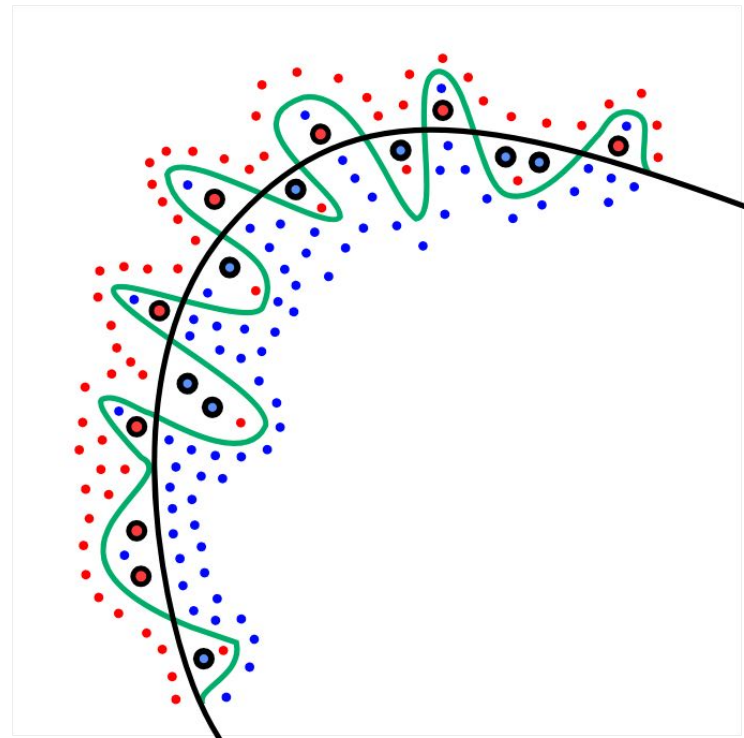
Regularized Cox Models

Need for Regularization:

- In high-dimensional datasets, traditional Cox models may **overfit**.
- Regularization introduces a penalty to reduce model complexity and enhance generalization.

Regularization (REVIEW):

- Lasso (L1);
- Ridge (L2);
- ElasticNet



Regularized Cox Models: In Practice

```
from sksurv.linear_model import CoxnetSurvivalAnalysis

# Create a pipeline that includes preprocessing and the CoxNetSurvivalAnalysis model
rcm = CoxnetSurvivalAnalysis(
    l1_ratio=0.5,          # Balance between L1 and L2 regularization
    alpha_min_ratio=0.01, # Smallest alpha as a fraction of the maximum alpha
    n_alphas=100,         # Number of alpha values to try
    max_iter=1000,        # Maximum number of iterations
    tol=1e-7,             # Tolerance for the optimization algorithm
    selection='cyclic'    # Method for selecting the active set of coefficients
)

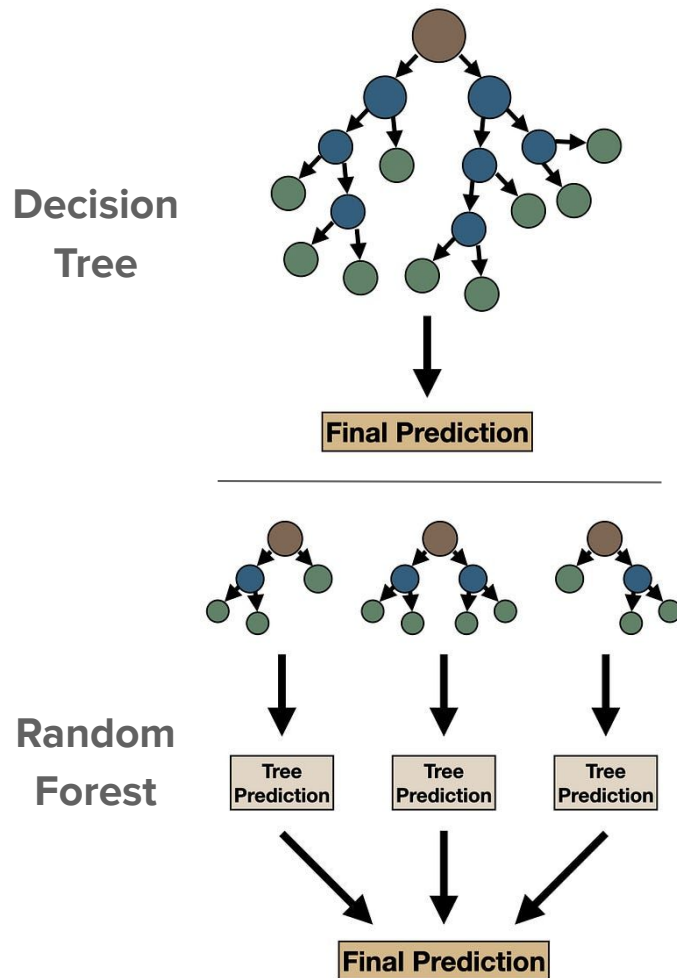
# Fit the model to the training data
pipeline.fit(X_train, y_train)
```

Random Forests (Review)

Overview: Ensemble-based model utilizing decision trees.

Mechanism:

- Constructs trees using bootstrapped samples.
- Determines optimal splits based on specified criteria at each node.
- Aggregates predictions through ensemble voting.

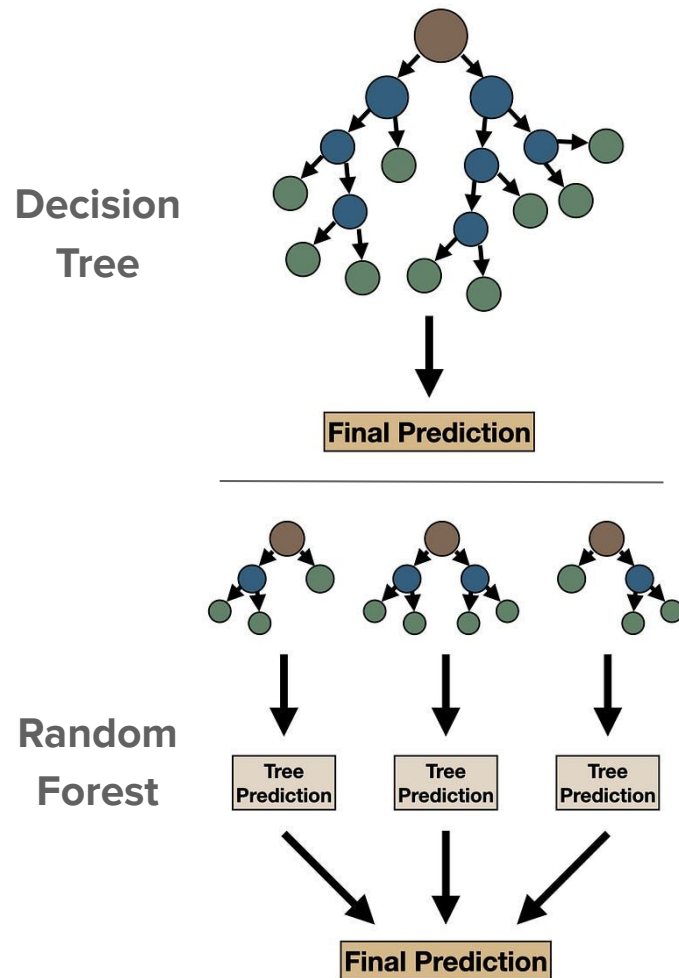


Random **Survival** Forests

Overview: Ensemble-based **survival** model utilizing decision trees.

Mechanism:

- Constructs trees using bootstrapped samples.
- Determines optimal splits based on **survival differences** at each node.
- Aggregates predictions through ensemble voting.



Random Survival Forests: In Practice

```
from sksurv.ensemble import RandomSurvivalForest

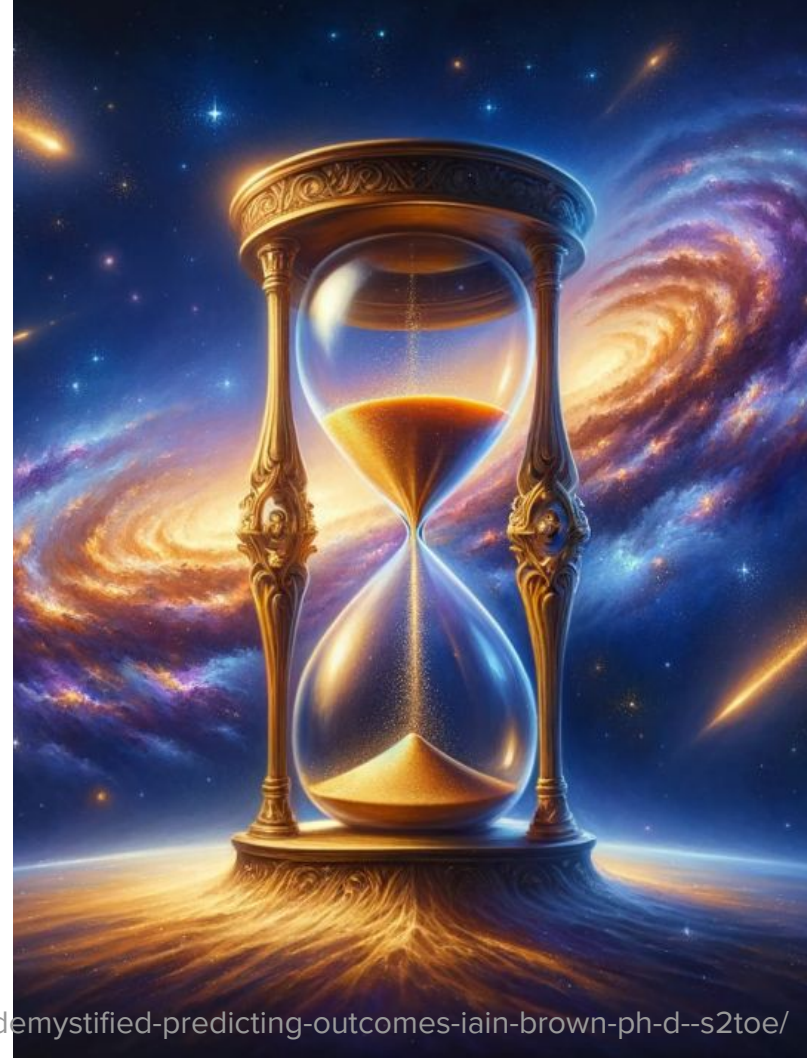
# Instantiate the RandomSurvivalForest model
rsf = RandomSurvivalForest(
    n_estimators=100,          # Number of trees in the forest
    min_samples_split=10,     # Minimum number of samples required to split an internal node
    min_samples_leaf=15,      # Minimum number of samples required to be at a leaf node
    max_features="sqrt",      # Number of features to consider when looking for the best split
    n_jobs=-1,                # Number of jobs to run in parallel (-1 means using all processors)
    random_state=42           # Seed used by the random number generator
)

# Fit the model to the training data
rsf.fit(X_train, y_train)
```

Use Cases

Survival Analysis Use Cases

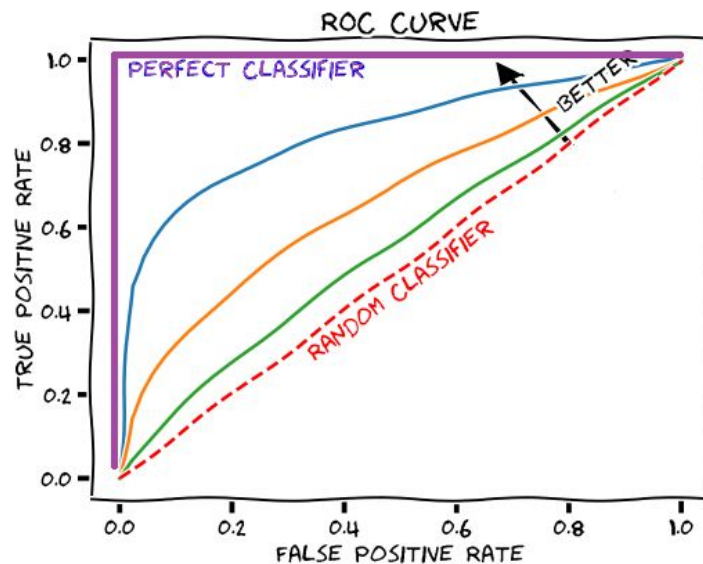
- **Clinical bioinformatics applications:**
 - Predicting patient survival using genomic data.
 - Risk stratification based on molecular markers.
 - Evaluating drug efficacy in clinical trials.
- **Beyond healthcare:**
 - **Engineering:** Predicting failure times of mechanical components.
 - **Business analytics:** Customer churn prediction.



Metrics for Evaluation

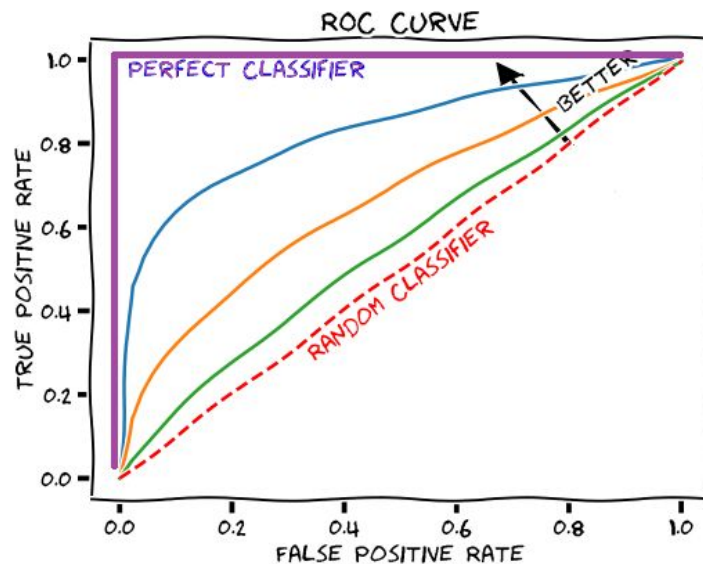
Concordance Index (C-Index)

- Generalization of the area under the ROC curve (AUROC);



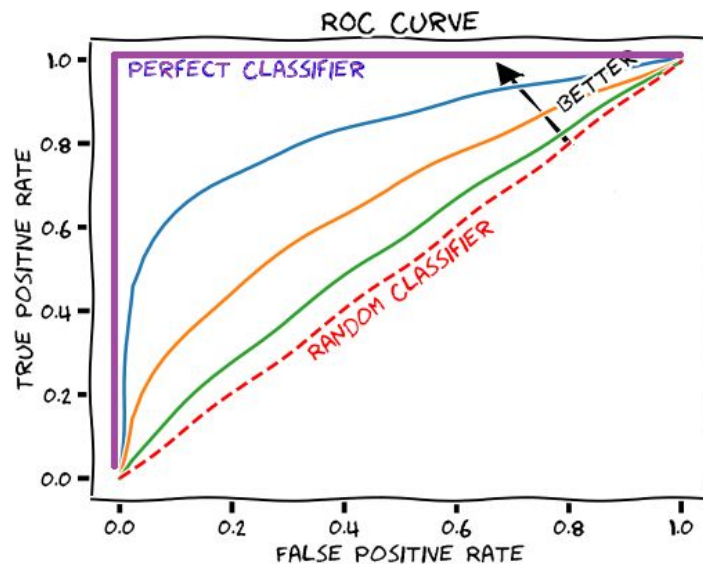
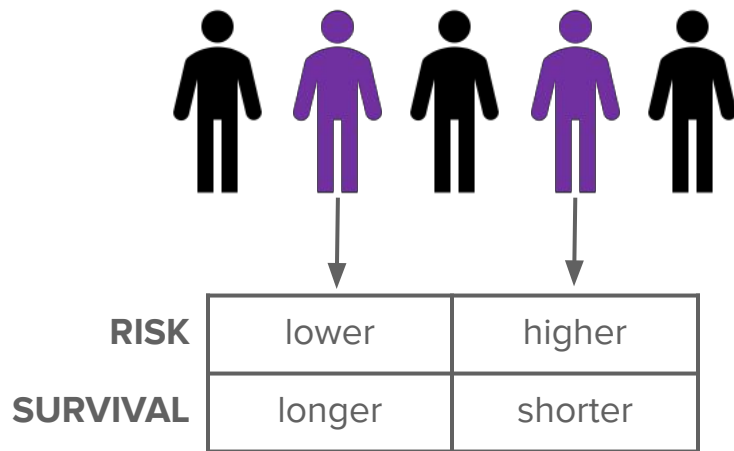
Concordance Index (C-Index)

- Generalization of the area under the ROC curve (AUROC);
- Concordant pairs:



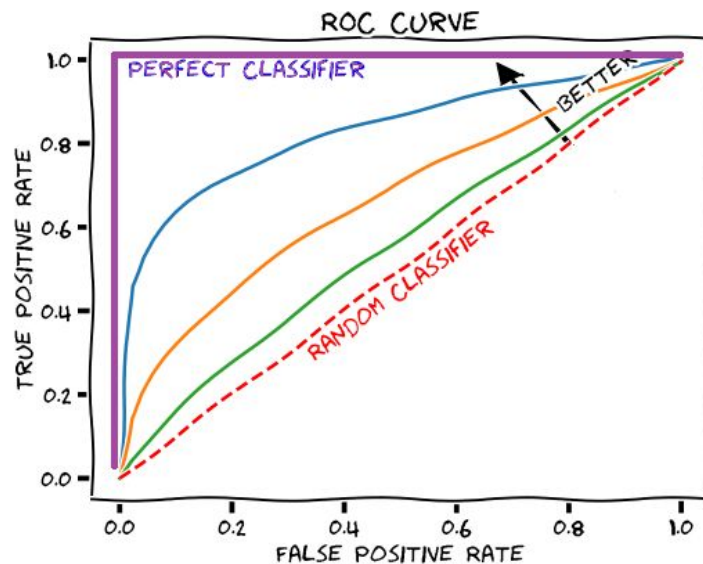
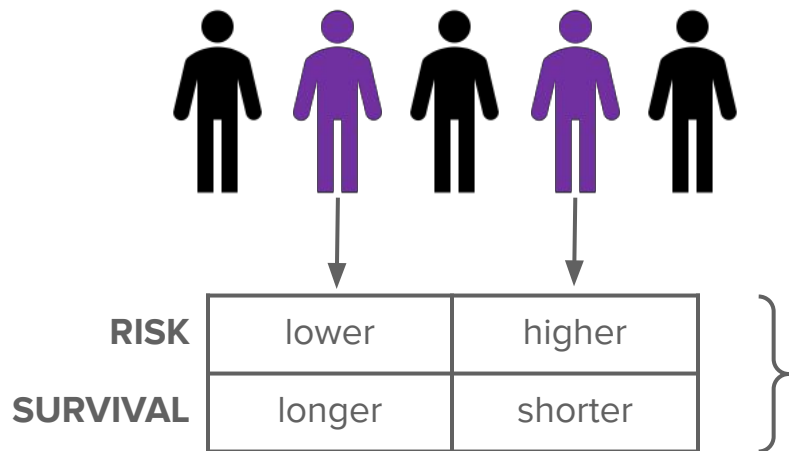
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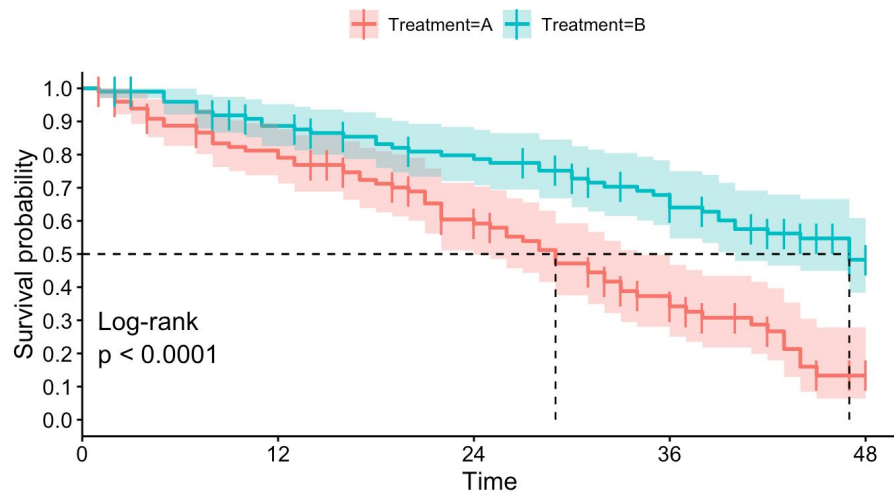


$$\text{C-Index} = \frac{\# \text{ concordant pairs}}{\text{total pairs}}$$

Other Metrics

- **Log-Rank Test:**

- Non-parametric test assessing the null hypothesis that groups have identical survival functions.



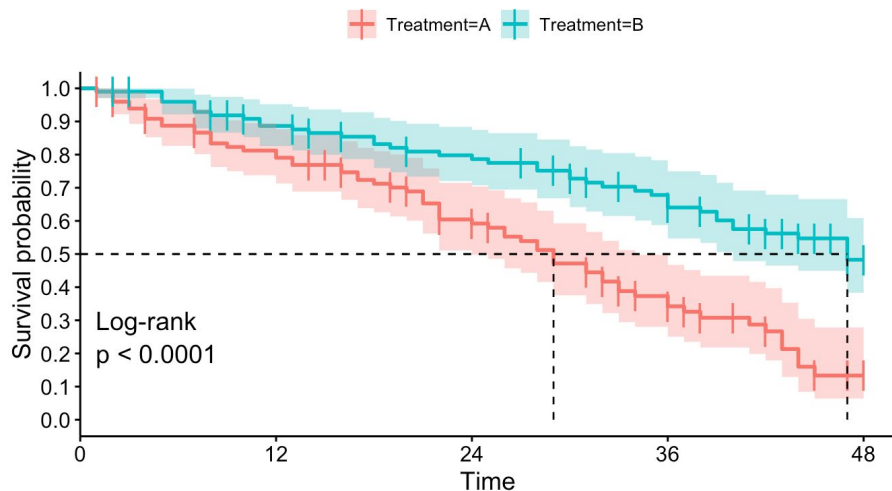
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- Non-parametric test assessing the null hypothesis that groups have identical survival functions.

- **Akaike Information Criterion (AIC):**

- Assesses the relative quality of statistical models for a given dataset.
- Balances model fit and complexity; lower AIC indicates a better model.
- Useful for model selection in survival analysis.



Challenges and Summary

Challenges in Survival Analysis

Challenge	Approach	Description
Non-Proportional Hazards	Stratified Cox Models	Allows different baseline hazards for strata defined by covariates.
	Time-Varying Covariates	Models covariates whose effects change over time.
	Alternative Models	Utilizes flexible parametric or machine learning models to capture complex relationships.

Challenges in Survival Analysis

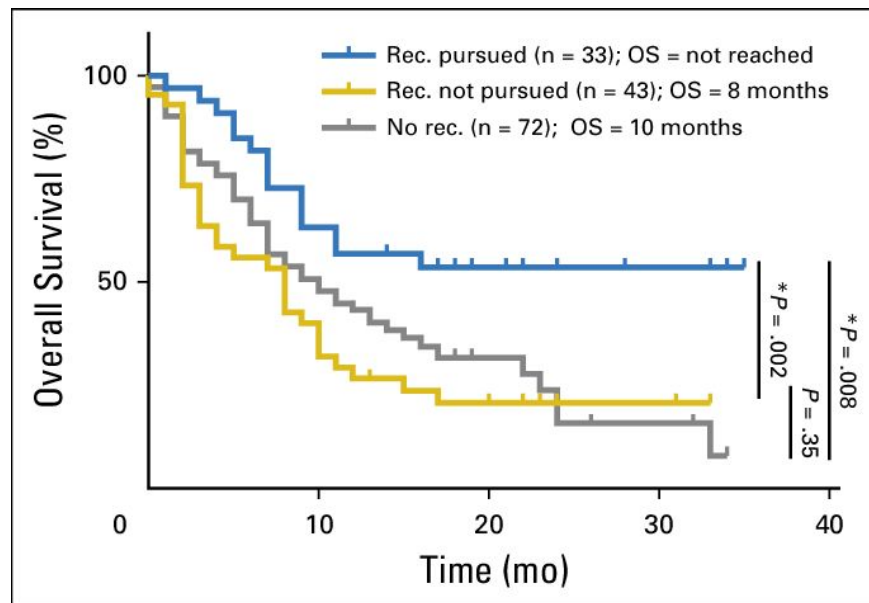
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Competing Risks	Cause-Specific Hazard Models	Analyzes the hazard of a specific event, treating other events as censored.
	Subdistribution Hazard Models	Evaluates the probability of a specific event occurring in the presence of competing events.
	Cumulative Incidence Functions	Estimates the probability of each event type over time, accounting for competing risks.

Challenges in Survival Analysis

Challenge	Approach	Description
Complex Datasets Interpretation	Validating Model Assumptions	Uses diagnostic plots to ensure data meet model assumptions.
	Managing High-Dimensional Data	Applies feature selection or penalized regression to handle numerous variables and prevent overfitting.

Key Takeaways

- **Survival analysis** is crucial for **time-to-event data**.
- **Kaplan-Meier** and **Cox models** are fundamental tools.
- **Regularized Cox regression** and **Random Survival Forests** are more advanced techniques;
- Metrics like **concordance index** and **log-rank test** ensure accurate evaluation.



Questions?



Logistic
regression
for
censored data



Cox
regression for
censored data